

PREDICTION OF RIGID WHEEL PERFORMANCE BASED ON THE ANALYSIS OF SOIL–WHEEL STRESSES PART I. PERFORMANCE OF DRIVEN RIGID WHEELS

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INTRODUCTION

RECENT investigations by the authors have shown that a sound theoretical basis for the prediction of wheel performance on soil can only be provided by knowledge of the actual soil failure phenomena beneath wheels [1]. For a very wide wheel (e.g. a wide roller or possibly a terra-tyre), soil failure only occurs in the direction of motion and it is, therefore, a two dimensional problem. On the other hand existing theory for wheels assumes soil failure and flow to be almost entirely sideways. The investigations referred to above have shown however, that beneath a conventional rather narrow wheel soil fails in a composite way, partly sideways and partly longitudinal. The longitudinal failure phenomenon usually cannot be ignored even for a quite narrow wheel, and the failure pattern is therefore much more complicated than existing theory assumes.

A great deal of effort will be needed before a three dimensional failure pattern can be established and only when this has been completed can a sound theoretical analysis of the problem be attempted. However, recent investigations do indicate that the sideways soil failure still plays an important role beneath conventional wheels [1]. Therefore, as an expedient at the present stage of the development of soil–vehicle mechanics, a theory for predicting the performance of conventional wheels based on the plate penetration test but with considerations of the important aspects of the actual interaction between wheels and soil should be developed.

This paper describes such an approach considering the distribution of both radial and tangential stresses on the soil–wheel interface and taking into account that they not only depend on soil properties and wheel dimensions but also on the slip (or skid). A comprehensive procedure for predicting the performance of conventional wheels is produced. It is shown that this method can give better prediction of wheel performance than existing theory.

This work consists of two parts. Part I concentrates on the discussion of the performance of driven rigid wheels, particularly with reference to sand. Towed rigid wheels will be discussed in Part II.

1. THE DISTRIBUTION OF RADIAL AND TANGENTIAL STRESSES ON THE SOIL–WHEEL INTERFACE

As is well known, the equilibrium of a wheel, working on horizontal ground with constant velocity, can be described by the following equations:

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$$W = rb \left\{ \int_{\theta_2}^{\theta_1} \sigma(\theta) \cos \theta \, d\theta + \int_{\theta_2}^{\theta_1} \tau(\theta) \sin \theta \, d\theta \right\}, \quad (1)$$

$$D = rb \left\{ \int_{\theta_2}^{\theta_1} \tau(\theta) \cos \theta \, d\theta - \int_{\theta_2}^{\theta_1} \sigma(\theta) \sin \theta \, d\theta \right\}, \quad (2)$$

$$T = r^2 b \int_{\theta_2}^{\theta_1} \tau(\theta) \, d\theta. \quad (3)$$

Where $\sigma(\theta)$ and $\tau(\theta)$ represent the average radial and tangential stress across the wheel width. Figure 1 illustrates the forces, torque and stresses acting on a wheel.

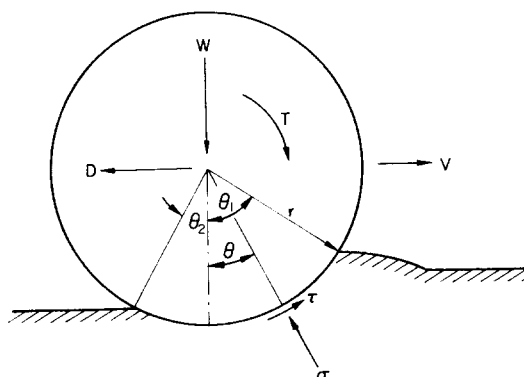


FIG. 1 Forces, torque and stresses acting on a driven rigid wheel.

It is therefore necessary to determine the relationships between radial and tangential stresses and soil properties, wheel dimensions and the degree of slip, in order to predict the wheel performance.

1.1 Radial stress distribution

Existing theory assumes that the radial soil reaction at an element of the rim is the same as that beneath a plate penetrated to the same depth [2]. This implies that the basic characteristics of the distribution of radial stress and motion resistance are independent of slip. Experiments on clay by Uffelmann show that the radial stress distribution seems to be quite uniform and quite independent of slip [3]. But a number of experiments on sand have shown that the radial stress distribution is a function of slip and the maximum radial stress does not act at the point beneath the axle ("bottom-dead-centre") as would be expected from the plate sinkage analogy. It actually occurs in front of bottom-dead-centre and shifts forward with increasing slip [4-7]. This indicates that special attention should be

paid to analysing the characteristics of radial stress distribution on sand and the existing theory should be modified accordingly.

The authors have previously shown that the maximum radial stress occurs at the point where the two soil failure zones beneath the wheel join each other, as at point *A* shown in Fig. 2 [1]. But it is not possible to locate this point for a driven

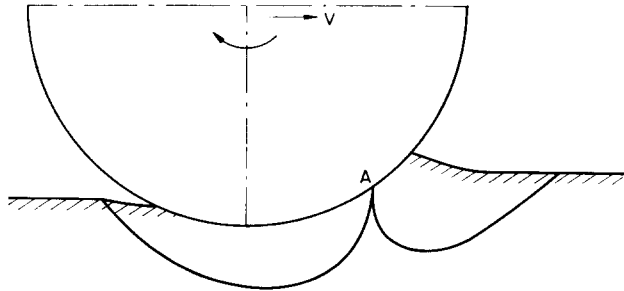


FIG. 2. Diagram showing the two failure zones beneath a driven rigid wheel on the longitudinal plane.

wheel by a theoretical approach at the moment, owing to the difficulty in establishing the three dimensional failure pattern. However, several sets of experimental results are now available [4-7] and these show that the location of the point of the maximum radial stress can be expressed by the following equation, when the wheel is working on sand :

$$\frac{\theta_M}{\theta_1} = c_1 + c_2 i. \quad (4)$$

Figure 3 shows the experimental results for the relative position of the maximum radial stress for different wheels under different working conditions [4-7].

The coefficients c_1 and c_2 are tabulated in Table 1.

TABLE 1

Soil	Angle of internal shearing resistance ϕ	Soil cohesion c (lb/in. ²)	Density γ (lb/in. ³)	Coefficients for determining the relative position of maximum radial stress		Remarks
				c_1	c_2	
Compact sand	33.3°	0.10	0.0575	0.43	0.32	According to Onafeko
Loose sand	31.1°	0.12	0.048	0.18	0.32	According to Hegedus
Sand	36.0°	0.10	0.0617	0.285	0.32	According to Sela
Dry sand	24.0°	—	—	0.38	0.41	

It is interesting to note that even though the experimental conditions are various, the coefficient c_2 has almost the same value, only c_1 is different. This implies that the relationship between the relative position of the maximum radial stress and slip has the same basic characteristic, that is to say the characteristic lines [from equation (4)] have almost the same slope. The only difference between them is

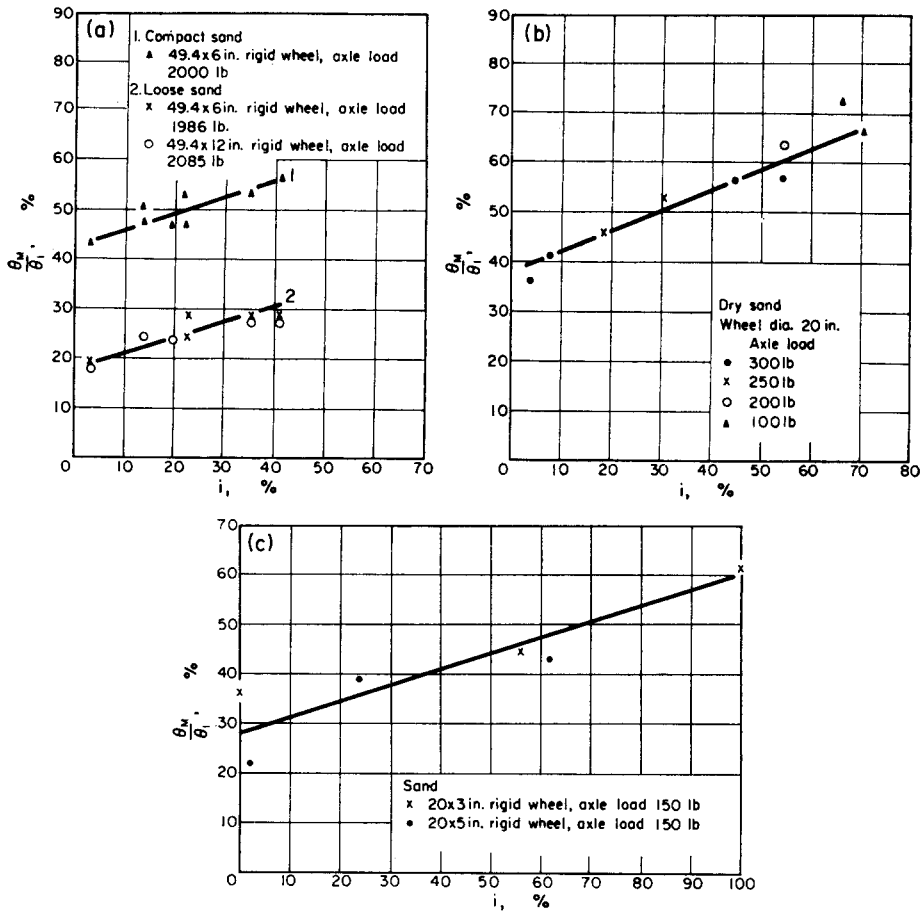


FIG. 3. Variation of the relative position of the maximum radial stress point with slip.

- (a) 49.4x6 in. and 49.4x12 in. rigid wheels on compact and loose sand (according to Onafeko [4]).
 (b) 20 in. dia. rigid wheel on dry sand (according to Sela [6]).
 (c) 20x5 in. and 20x3 in. rigid wheels on sand (according to Hegedus [7]).

in the initial value.

It therefore seems that the relative position of the maximum radial stress depends on the compressibility of the sand, and it is quite independent of wheel dimensions and sinkage. Of course, more theoretical and experimental investigations are needed before such a comprehensive conclusion can be completely justified.

Experimental evidence [4], [5] has also shown that on sand from the beginning of contact θ_1 to the maximum radial stress point θ_M (Fig. 4), the radial stress around the rim can be quite well described by the pressure-sinkage relationship proposed by Reece [8]. In other words, in the region between θ_1 and θ_M (called the "front region" and illustrated in Fig. 4), the radial stress can be calculated by the following equation:

$$\sigma_1 = (k_1 + k_2 b) \left(\frac{z}{b} \right)^n. \quad (5)$$

If the relationship $z = (\cos \theta - \cos \theta_1) r$ is used, then equation (5) can be re-written as follows:

$$\sigma_1(\theta) = (k_1 + k_2 b) \left(\frac{r}{b} \right)^n (\cos \theta - \cos \theta_1)^n. \quad (6)$$

From the maximum radial pressure point θ_M to the end of contact θ_2 (called the "rear region" and illustrated in Fig. 4), the radial stress around the rim decreases. But it is interesting to note that the radial stress distribution still has similar characteristics. This suggests that if the relative position to the corresponding free surface of a point in the rear region is the same as that of one in the front region, then both will have the same value of radial stress. This can be described by the

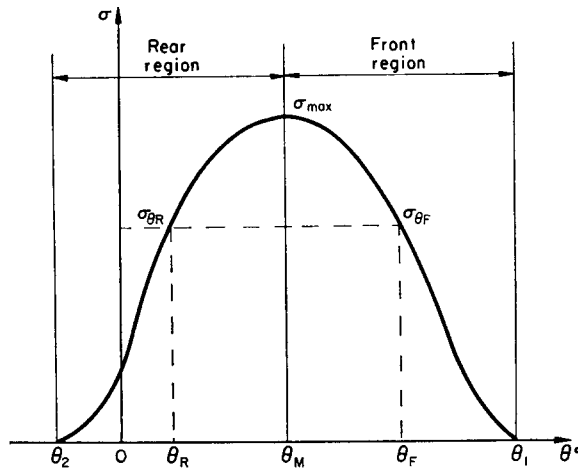


FIG. 4. Diagram illustrating the characteristic distribution of radial stress on the soil-wheel interface of a driven rigid wheel.

following equations and is illustrated in Fig. 4, i.e.

$$\frac{\theta_R - \theta_2}{\theta_M - \theta_2} = \frac{\theta_1 - \theta_F}{\theta_1 - \theta_M}, \quad (7)$$

then

$$\sigma_{\theta R} = \sigma_{\theta F}. \quad (8)$$

According to the above proposal, the radial stress acting in the rear region can be calculated by the following equation:

$$\sigma_2(\theta) = (k_1 + k_2 b) \left(\frac{r}{b} \right)^n \left[\cos \left(\theta_1 - \left(\frac{\theta - \theta_2}{\theta_M - \theta_2} \right) (\theta_1 - \theta_M) \right) - \cos \theta_1 \right]^n \quad (9)$$

θ_2 represents the degree of rut recovery. It is quite clear that rut recovery is a function of slip and wheel width [1], but the quantitative relationship between them has not yet been established. If rut recovery is small and neglected (e.g. the wheel is not wide and the degree of slip is not very high), the above equation can

be simplified as follows :

$$\sigma_2(\theta) = (k_1 + k_2 b) \left(\frac{r}{b} \right)^n \left[\cos \left(\theta_1 - \frac{\theta}{\theta_M} (\theta_1 - \theta_M) \right) - \cos \theta_1 \right]^n. \quad (10)$$

According to equation (4), θ_M can be replaced by the following equation :

$$\theta_M = (c_1 + c_2 i) \theta_1.$$

Equation (10) can then be rewritten as follows :

$$\sigma_2(\theta) = (k_1 + k_2 b) \left(\frac{r}{b} \right)^n \left[\cos \left(\theta_1 - \theta \left(\frac{1 - (c_1 + c_2 i)}{(c_1 + c_2 i)} \right) \right) - \cos \theta_1 \right]^n. \quad (11)$$

From the foregoing analysis, it is clear that if the pressure-sinkage constants k_1 , k_2 and n , the coefficients c_1 and c_2 , the wheel dimensions r and b , sinkage z and slip i are known, then using equations (4), (6) and (9) or (11), the radial stress distribution on the soil-wheel interface can be determined quantitatively.

Figures 5 and 6 show that this can give a satisfactory description of the radial stress distribution around a driven rigid wheel working on different sands. The predicted distribution is obtained from the measured sinkage using equations (4),

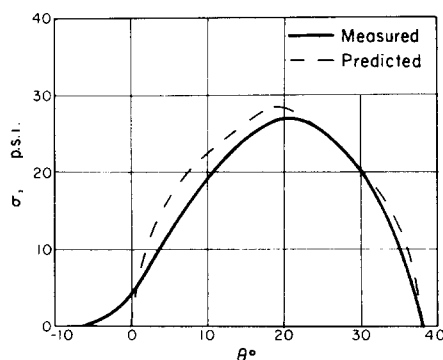


FIG. 5. Comparison of the measured and predicted distribution of radial stress of a 49.4 × 6 in. rigid wheel at 22.1 per cent slip with the same sinkage on compact sand (measured values from Onafeko's experiments, soil values as in Table 2).

TABLE 2

Soil	Soil properties			Pressure-sinkage constants			Shear deformation modulus K (in.)	Coefficients for determining the relative position of maximum radial stress	
	Angle of internal shearing resistance ϕ	Soil cohesion c (lb/in. ²)	Density γ (lb/in. ³)	k_1	k_2	n		c_1	c_2
Compact sand	33.3°	0.10	0.0575	20	2.5	0.4706	1.5	0.43	0.32
Loose sand	31.1°	0.12	0.048	0	2	1.1504	1.5	0.18	0.32

(6) and (11), neglecting the effect of rut recovery. Soil properties and other constants used in the computation are tabulated in Table 2.

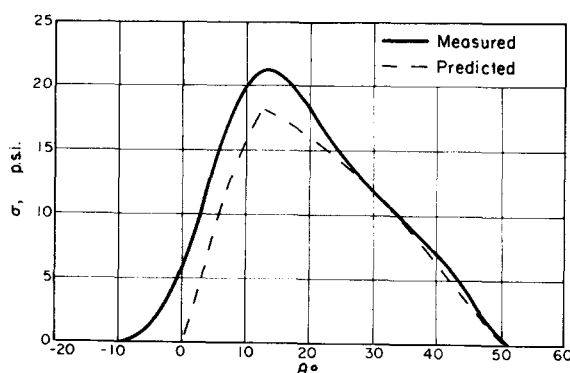


FIG. 6. Comparison of the measured and predicted distribution of radial stress of a 49.4×6 in. rigid wheel at 22.1 per cent slip with the same sinkage on loose sand (measured values from Onafeko's experiments, soil values as in Table 2).

1.2 Tangential stress distribution

It has been shown that shear stress developed in soil depends on shear deformation (displacement), and the relationship between them can be quite well described by the following simplified Bekker equation:

$$\tau = (c + \sigma \tan \phi) (1 - e^{-j/K}).$$

In order to evaluate the shear (tangential) stress distribution around the rim, it is, therefore, necessary to find out the shear deformation j along the soil-wheel interface.

The shear deformation beneath a track has been investigated by Bekker based on the analysis of the slip velocity v_j of the track and is illustrated in Fig. 7(a).

Following a similar approach, the shear deformation on the soil-wheel interface can be found. Beneath a driven rigid wheel, the slip velocity v_j of the rim relative to the soil is the tangential component of the absolute velocity v_a at the same point, as illustrated in Fig. 7(b). The slip velocity v_j can, therefore, be expressed by the equation:

$$v_j = r\omega [1 - (1 - i) \cos \theta].$$

The shear deformation j along the soil-wheel interface can then be calculated in the following way:

$$\begin{aligned} j &= \int_0^t v_j dt = \int_{\theta}^{\theta_1} r\omega [1 - (1 - i) \cos \theta] \frac{d\theta}{\omega} \\ &= r[(\theta_1 - \theta) - (1 - i)(\sin \theta_1 - \sin \theta)]. \end{aligned} \quad (12)$$

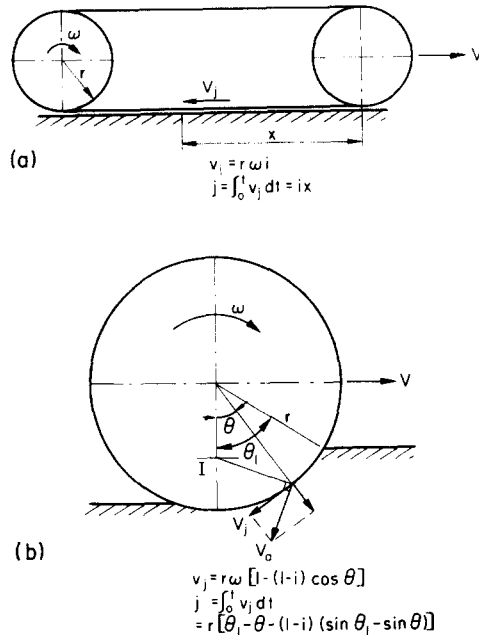


FIG. 7. Diagram illustrating the development of shear deformation beneath (a) a track and (b) a rigid wheel.

The shear stress around the rim can, therefore, be expressed by the following equation:

$$\begin{aligned} \tau(\theta) &= (c + \sigma(\theta) \tan \phi) (1 - e^{-j/K}) \\ &= (c + \sigma(\theta) \tan \phi) (1 - e^{-r/K [(\theta_1 - \theta) - (1-i)(\sin \theta_1 - \sin \theta)]}) \end{aligned} \quad (13)$$

In the front and rear regions, $\sigma(\theta)$ can be determined by equations (6) and (9) or (11) respectively.

Incidentally, the shear deformation along the soil-wheel interface has also been investigated by Onafeko using a different approach and the same result has been obtained [4, 5].

In order to carry out calculation of tangential stress from equation (13), the shear deformation modulus K should be determined beforehand. Experiments by Wills [9] and the authors have shown that the shear deformation modulus K seems to depend on normal stress. But the quantitative relationship between them still needs to be determined by further investigations. In this analysis, an average value of K is taken in the computation for a particular soil.

It has been shown that equation (13) can give fairly good prediction of tangential stresses around the rim of a driven wheel, particularly at high slip and on compact sand [4, 5]. But it needs some modification before it can be applied to predict the tangential stresses on the soil-wheel interface of a towed wheel, and this will be discussed in Part II of this paper.

2. PROCEDURE FOR PREDICTING THE PERFORMANCE OF DRIVEN RIGID WHEELS

If soil values c , ϕ , K , k_1 , k_2 , n , wheel dimensions r , b , the coefficients for determining the relative position of the maximum radial stress c_1 , c_2 and vertical axle load of the wheel W are known, the wheel performance can be predicted by the following procedure.

2.1 Prediction of sinkage

If rut recovery is neglected, by using equations (4), (6), (11) and (13), the equilibrium equation (1) can be rewritten as follows:

$$\begin{aligned}
 W = rb & \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} \sigma_1(\theta) \cos \theta d\theta + \int_0^{(c_1+c_2i)\theta_1} \sigma_2(\theta) \cos \theta d\theta \right. \\
 & + \left. \int_{(c_1+c_2i)\theta_1}^{\theta_1} \tau_1(\theta) \sin \theta d\theta + \int_0^{(c_1+c_2i)\theta_1} \tau_2(\theta) \sin \theta d\theta \right\} \\
 = rb & \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} (k_1 + k_2b) \left(\frac{r}{b} \right)^n (\cos \theta - \cos \theta_1)^n \cos \theta d\theta \right. \\
 & + \int_0^{(c_1+c_2i)\theta_1} (k_1 + k_2b) \left(\frac{r}{b} \right)^n \left(\cos \left(\theta_1 - \theta \left(\frac{1 - (c_1 + c_2i)}{c_1 + c_2i} \right) \right) - \cos \theta_1 \right)^n \cos \theta d\theta \\
 & + \int_{(c_1+c_2i)\theta_1}^{\theta_1} \left[c + (k_1 + k_2b) \left(\frac{r}{b} \right)^n (\cos \theta - \cos \theta_1)^n \tan \phi \right] \\
 & \times (1 - e^{-r/K [(\theta_1 - \theta) - (1-i)(\sin \theta_1 - \sin \theta)])} \sin \theta d\theta \\
 & + \int_0^{(c_1+c_2i)\theta_1} \left[c + (k_1 + k_2b) \left(\frac{r}{b} \right)^n \left(\cos \left(\theta_1 - \theta \left(\frac{1 - (c_1 + c_2i)}{c_1 + c_2i} \right) \right) - \cos \theta_1 \right)^n \tan \phi \right] \\
 & \times (1 - e^{-r/K [(\theta_1 - \theta) - (1-i)(\sin \theta_1 - \sin \theta)])} \sin \theta d\theta \left. \right\}. \quad (14)
 \end{aligned}$$

When W is given and other constants are known, the contact angle θ_1 can be found from the above equation (14), and the wheel sinkage z_0 can then be determined from

$$z_0 = (1 - \cos \theta_1)r. \quad (15)$$

Unfortunately, equation (14) is too complex to be integrated by the normal method. Therefore, in practice, Simpson's rule is used to evaluate the integrals and successive approximation method is then applied to find out the contact angle. This process can easily be carried out on a high speed digital computer.

It is interesting to point out that by using equation (14) to predict the wheel age, the effect of supporting part of the axle load by the vertical components of tangential stresses is considered. This is neglected in existing theory when predicting sinkage (2).

Prediction of drawbar pull and torque input

When the contact angle θ_1 has been determined by equation (14), the drawbar D can be predicted by the following equation:

$$\begin{aligned}
 D = & rb \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} \tau_1(\theta) \cos \theta d\theta + \int_0^{(c_1+c_2i)\theta_1} \tau_2(\theta) \cos \theta d\theta \right\} \\
 & - rb \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} \sigma_1(\theta) \sin \theta d\theta + \int_0^{(c_1+c_2i)\theta_1} \sigma_2(\theta) \sin \theta d\theta \right\} \\
 = & rb \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} \left[c + (k_1 + k_2b) \left(\frac{r}{b} \right)^n (\cos \theta - \cos \theta_1)^n \tan \phi \right] \right. \\
 & \times (1 - e^{-r/K [(\theta_1 - \theta) - (1-i)(\sin \theta_1 - \sin \theta)])} \cos \theta d\theta \\
 & + \int_0^{(c_1+c_2i)\theta_1} \left[c + (k_1 + k_2b) \left(\frac{r}{b} \right)^n \left(\cos \left(\theta_1 - \theta \left(\frac{1 - (c_1 + c_2i)}{c_1 + c_2i} \right) \right) - \cos \theta_1 \right)^n \tan \phi \right] \\
 & \times (1 - e^{-r/K [(\theta_1 - \theta) - (1-i)(\sin \theta_1 - \sin \theta)])} \cos \theta d\theta \Big\} \\
 & - rb \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} (k_1 + k_2b) \left(\frac{r}{b} \right)^n (\cos \theta - \cos \theta_1)^n \sin \theta d\theta \right. \\
 & + \int_0^{(c_1+c_2i)\theta_1} (k_1 + k_2b) \left(\frac{r}{b} \right)^n \left(\cos \left(\theta_1 - \theta \left(\frac{1 - (c_1 + c_2i)}{c_1 + c_2i} \right) \right) - \cos \theta_1 \right)^n \sin \theta d\theta \Big\}.
 \end{aligned} \tag{16}$$

The first term on the right hand side of equation (16) is usually defined as thrust H and the last term as motion resistance R . This indicates that not only thrust but also motion resistance is a function of slip.

The torque T is similarly obtained from :

$$\begin{aligned}
 T &= r^2 b \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} \tau_1(\theta) d\theta + \int_0^{(c_1+c_2i)\theta_1} \tau_2(\theta) d\theta \right\} \\
 &= r^2 b \left\{ \int_{(c_1+c_2i)\theta_1}^{\theta_1} \left[c + (k_1 + k_2 b) \left(\frac{r}{b} \right)^n (\cos \theta - \cos \theta_1)^n \tan \phi \right] \right. \\
 &\quad \times (1 - e^{-r/K [(\theta_1 - \theta) - (1-i) (\sin \theta_1 - \sin \theta)])} d\theta \\
 &\quad + \int_0^{(c_1+c_2i)\theta_1} \left[c + (k_1 + k_2 b) \left(\frac{r}{b} \right)^n \left(\cos \left(\theta_1 - \theta \left(\frac{1 - (c_1 + c_2 i)}{c_1 + c_2 i} \right) \right) - \cos \theta_1 \right)^n \tan \phi \right] \\
 &\quad \times (1 - e^{-r/K [(\theta_1 - \theta) - (1-i) (\sin \theta_1 - \sin \theta)])} d\theta \left. \right\}. \quad (17)
 \end{aligned}$$

Again, Simpson's rule is used to calculate the integrals in both equations (16) and (17) on a digital computer.

A block scheme illustrating the computer program for predicting the performance of driven rigid wheels using the above procedure is given in the appendix.

The comparison of the measured and predicted performance of two rigid wheels on different sands using this computer program is shown in Figs. 8, 9 and 10. Soil properties and other constants used in the computation are shown in Table 2.

As examples, Figs. 11, 12 and 13 show the comparison of the measured and predicted radial and tangential stresses distribution on the soil-wheel interface of driven wheels under different conditions.

It is shown that within the experimental range there is a quite close agreement between the measured and predicted performance, particularly on compact sand.

It is hoped that the procedure described can also be applied to predict the performance of rigid wheels on clay, if the coefficients in the equations are substituted with the appropriate ones for clay. However, this needs further investigation.

CONCLUSIONS

1. A comprehensive frame work for predicting the performance of driven rigid wheels has been established. All known major factors affecting driven wheel performance, especially the effect of slip on sinkage, motion resistance and thrust, have been taken into account. It is shown that the method described can give better prediction than existing theory.

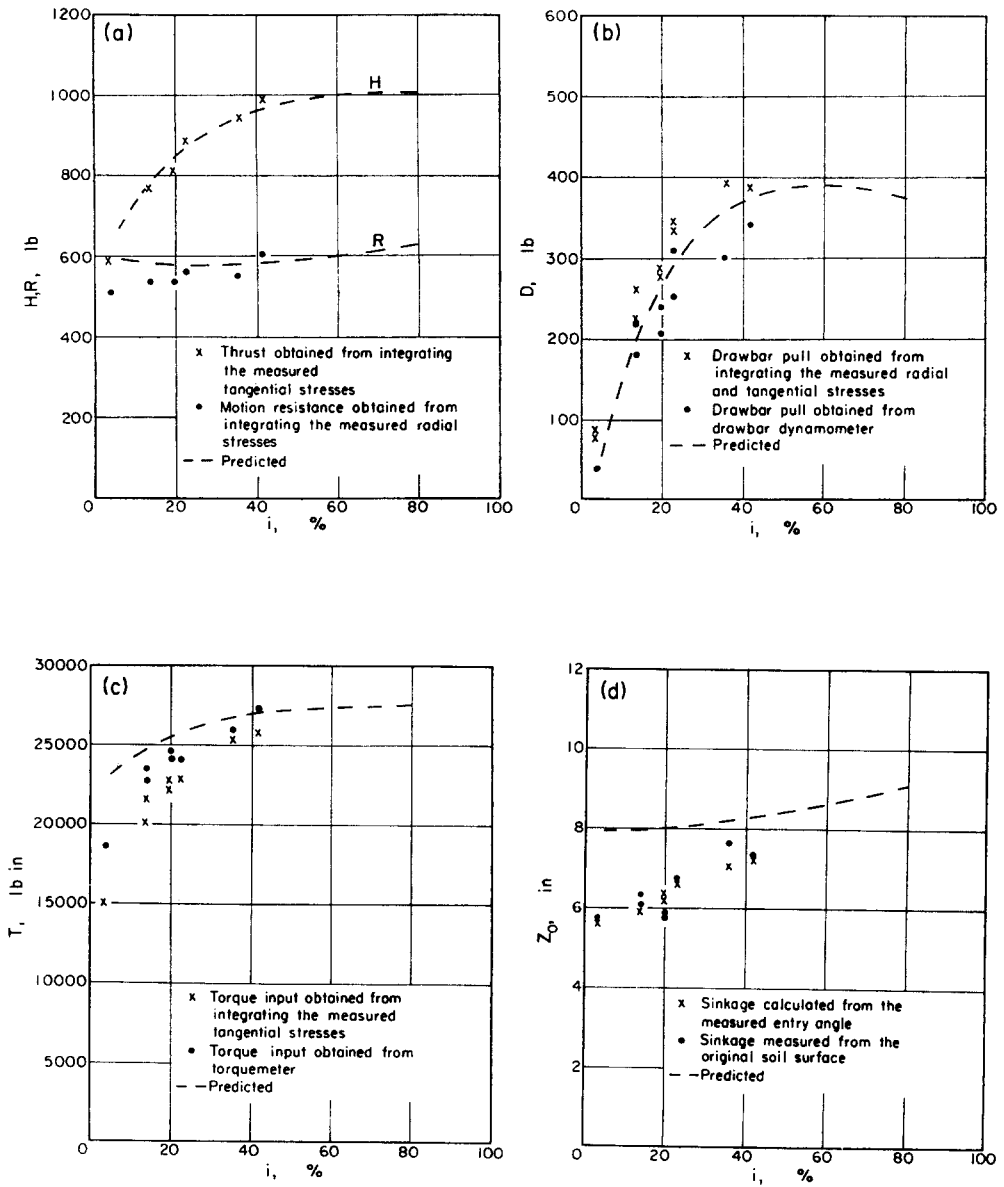


Fig. 8. Comparison of the measured and predicted performance of a 49.4x6 in. driven rigid wheel with axle load 2000 lb on compact sand (measured values from Onafeko's experiments).

- (a) thrust and motion resistance vs. slip;
- (b) drawbar pull vs. slip;
- (c) torque input vs. slip;
- (d) sinkage vs. slip.

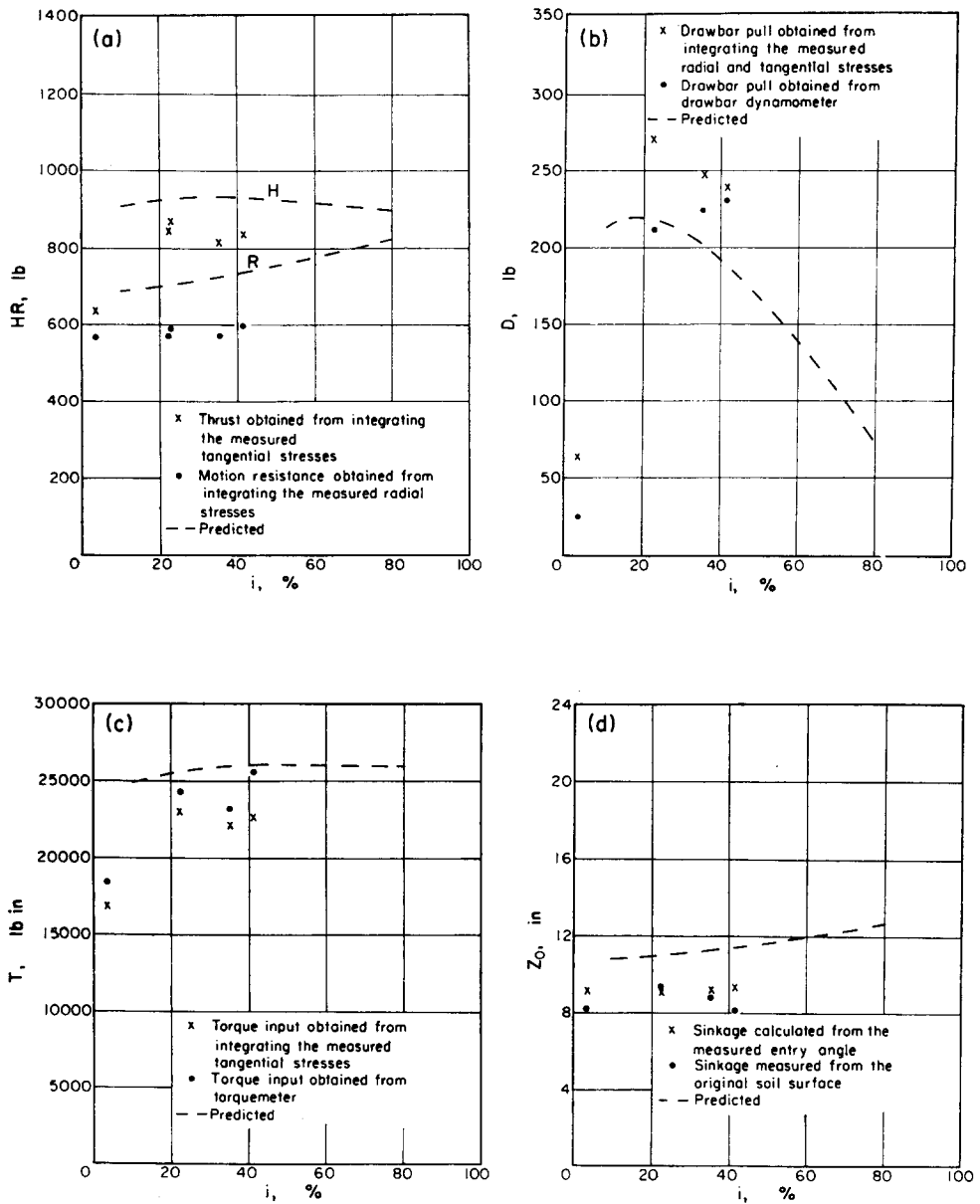


FIG. 9. Comparison of the measured and predicted performance of a 49.4x6 in. driven rigid wheel with axle load 1986 lb on loose sand (measured values from Onafeko's experiments).

- (a) thrust and motion resistance vs. slip;
- (b) drawbar pull vs. slip;
- (c) torque input vs. slip;
- (d) sinkage vs. slip.

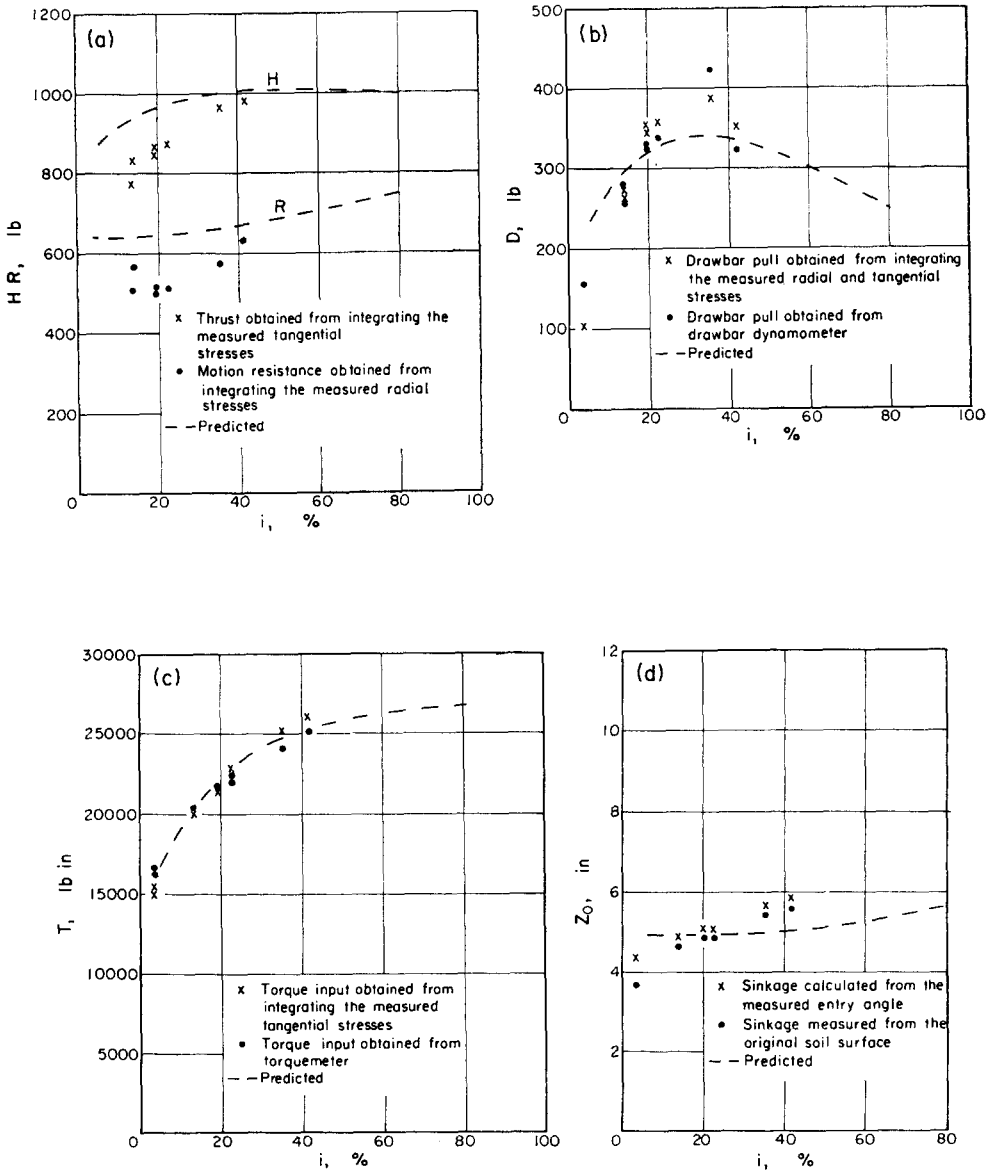


FIG. 10. Comparison of the measured and predicted performance of a 49.4 x 12 in. driven rigid wheel with axle load 2085 lb on loose sand (measured values from Onafeko's experiments).

- (a) thrust and motion resistance vs. slip;
- (b) drawbar pull vs. slip;
- (c) torque input vs. slip;
- (d) sinkage vs. slip.

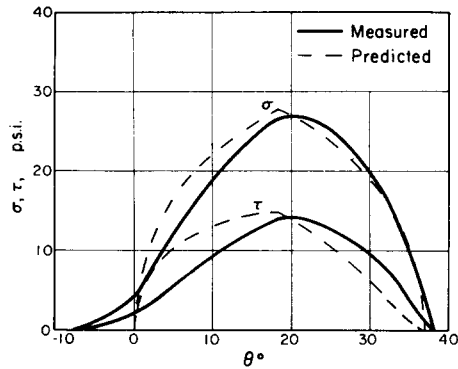


FIG. 11. Comparison of the measured and predicted distribution of the radial and tangential stresses of a 49.4×6 in. driven rigid wheel with axle load 2000 lb at 22.1 per cent slip on compact sand (measured values from Onafeko's experiments).

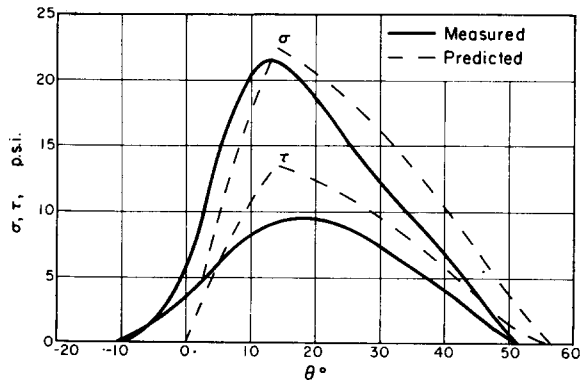


FIG. 12. Comparison of the measured and predicted distribution of the radial and tangential stresses of a 49.4×6 in. driven rigid wheel with axle load 1986 lb at 22.1 per cent slip on loose sand (measured values from Onafeko's experiments).

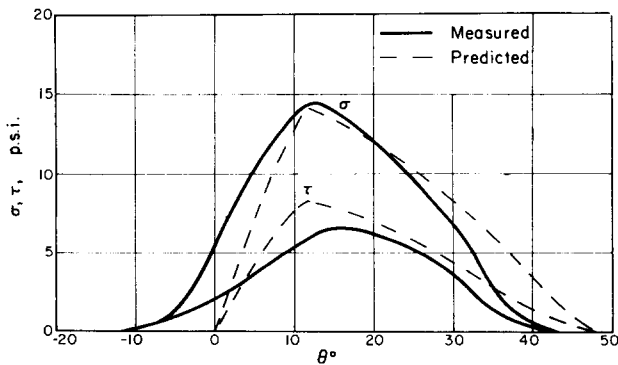


FIG. 13. Comparison of the measured and predicted distribution of the radial and tangential stresses of a 49.4×12 in. driven rigid wheel with axle load 2085 lb at 22.1 per cent slip on loose sand (measured values from Onafeko's experiments).

2. The relationship between the radial stress distribution on the soil-wheel interface and slip is described, taking into account the fact that the maximum radial stress shifts forward with increasing slip. This leads to the result that motion resistance and sinkage are functions of slip, which is consistent with the usually observed "slip-sinkage" phenomenon.

3. Based on the analysis of the relative movement between the rim and the soil, a theoretical expression to describe the shear deformation along the soil-wheel interface has been derived. It gives a satisfactory prediction of shear stress distribution at high slips.

4. The effect of supporting part of the axle load by the vertical components of tangential stresses is considered in the prediction of wheel sinkage. This gives more accurate results.

5. When using the method described to predict the performance of driven rigid wheels, the coefficients for locating the relative position of the maximum radial stress on the soil-wheel interface have to be determined by experiment. Although, it is believed that the empirical values of the coefficients given in the paper could probably cover quite a wide range of working conditions on sand, a general theoretical approach for locating the maximum radial stress point still needs to be developed in order to make a more comprehensive picture.

6. Before this theory can be accepted as generally true it will need more experimental support.

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NOTATION

b	wheel width
c	soil cohesion
c_1, c_2	coefficients for determining the relative position of the maximum radial stress on the soil-wheel interface
D	drawbar pull
H	thrust
i	slip
j	shear deformation along the soil-wheel interface
K	shear deformation modulus
k_1, k_2	pressure sinkage moduli
n	exponent of sinkage to width ratio
r	radius of the wheel
R	motion (rolling) resistance
T	torque input
V	forward velocity of the wheel axle
v_a	absolute velocity of a point on the rim
v_j	slip velocity of a point on the rim (the tangential component of the absolute velocity v_a)
W	vertical axle load of the wheel
z	sinkage
z_0	wheel sinkage
σ	radial stress on the soil-wheel interface
σ_1	radial stress in the front region of the soil-wheel interface
σ_2	radial stress in the rear region of the soil-wheel interface
τ	tangential (shear) stress on the soil-wheel interface
τ_1	tangential (shear) stress in the front region of the soil-wheel interface
τ_2	tangential (shear) stress in the rear region of the soil-wheel interface

- θ angle
 θ_1 entry angle (or contact angle when rut recovery is neglected)
 θ_2 exit angle
 θ_M angular position of the maximum radial stress
 ϕ angle of internal shearing resistance of soil
 ω angular velocity of the wheel

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APPENDIX

BLOCK SCHEME OF THE COMPUTER PROGRAMME FOR PREDICTING THE PERFORMANCE OF DRIVEN RIGID WHEELS

